

Rethinking the Utilization of Individual Rewards in Multiagent Reinforcement Learning With Sparse Team Rewards

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Abstract—The sparsity of team rewards significantly hinders the learning of optimal team policies in cooperative multiagent reinforcement learning (MARL). While augmenting sparse team rewards with individual rewards is a common solution, existing methods face three critical challenges: 1) inconsistency between the learned policy and the optimal team policy due to reward function modification; 2) incompatibility with different individual reward settings; and 3) suboptimal balance between individual and team reward-oriented policy optimization. To address these challenges, we propose CLOT, a novel policy consistency constrained multiagent policy optimization approach that leverages individual rewards in a reward setting-agnostic manner. Specifically, we first present a constrained policy optimization problem formulated by a consistency constraint between the team returns of the learned policy and those of the optimal team policy. Then, we develop a Lagrangian dual-based iterative policy optimization procedure to solve the formulated problem, deriving exact optimization objectives for policy training. Throughout this process, a dynamic Lagrangian multiplier update mechanism is proposed to automatically balance individual and team reward-oriented policy optimization. Extensive experimental evaluations across the StarCraft II Multiagent Challenge (SMAC), multiagent particle environment (MPE), and Google research football (GRF) environments demonstrate that our approach effectively addresses all three identified challenges, significantly enhancing performance in cooperative multiagent scenarios with sparse team rewards.

Index Terms—Multiagent reinforcement learning (MARL), policy consistency, reinforcement learning, sparse reward.

I. INTRODUCTION

COOPERATIVE multiagent reinforcement learning (MARL) has attracted significant interest due to its potential for solving complex decision-making tasks [1], [2]. The primary objective of cooperative MARL is to learn an optimal team policy π_E that maximizes the expected cumulative team rewards: $\pi_E^* = \arg \max_{\pi_E} \mathbb{E}_{\pi_E} [\sum_{t=0}^{\infty} \gamma^t r_t^E]$. However, sparse team rewards pose a significant challenge to achieving this objective [3], [4], [5]. A prominent example is the popular StarCraft II Multiagent Challenge (SMAC) environment, where agents often receive nonzero team rewards only at the conclusion of a battle based on victory or defeat outcomes, significantly hampering MARL algorithm performance [6], [7]. To address this challenge, a common approach involves introducing dense individual rewards alongside team rewards to guide agents in performing specific tasks. Continuing with the SMAC example, popular dense individual rewards employed by many MARL algorithms, such as multiagent proximal policy optimization (MAPPO) [8], are designed based on the health metrics of team members and enemies to encourage effective shooting or escaping action, which potentially contribute to winning the battle.

With the increasing prevalence of individual rewards in MARL, a significant question emerges: *how to optimally utilize the individual rewards in MARL?*

Based on the manner in which individual rewards are utilized, existing related studies can be primarily categorized into two categories: **reward shaping**-based approaches and **knowledge transfer**-based approaches. The former category [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21] employs reward shaping techniques [22] that combine team r_t^E and individual rewards r_t^i into a mixed reward function $\tilde{r}_t^i = r_t^i + r_t^E$ directly, as illustrated in Fig. 1(a). Reward shaping provides a potential approach for directly balancing team and individual rewards in mixed policy learning. However, existing approaches primarily focus on designing individual rewards through expert knowledge or systematic exploration of agents' state-action spaces. This overlooks a critical challenge: directly incorporating individual rewards fundamentally

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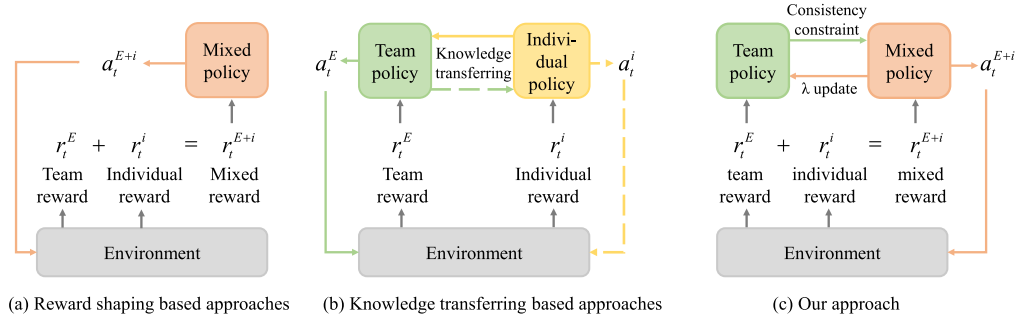


Fig. 1. Comparison of individual reward utilization in MARL. (a) Reward shaping-based approaches train a mixed policy using the direct sum of individual and team rewards, potentially causing deviation from optimal team policies. (b) Knowledge transfer approaches leverage insights from individual rewards to enhance team policy learning, with some methods (e.g., IRAT [23]) also transferring knowledge from team to individual policies (dashed lines). However, these approaches struggle when individual rewards conflict with team optimization. (c) CLOT simultaneously trains mixed and team policies with a consistency constraint, dynamically balancing team and individual rewards through Lagrangian multiplier (λ) updates, ensuring the mixed policy aligns with optimal team policy while benefiting from individual rewards.

changes each agent's optimization objective, so the product of individually optimal policies generally deviates from the team-optimal policy π_E^* , that is,

$$\prod_{i \in \mathcal{N}} \arg \max_{\pi_i} \mathbb{E}_{\pi_i} \left[\sum_{t=0}^{\infty} \gamma^t \hat{r}_t^i \right] \neq \arg \max_{\pi_E} \mathbb{E}_{\pi_E} \left[\sum_{t=0}^{\infty} \gamma^t r_t^E \right]. \quad (1)$$

Such policy inconsistency typically yields suboptimal team performance and emergent miscoordination: locally attractive actions (under $\hat{r}_t^i$) conflict across agents, amplifying small errors into joint failures. Furthermore, most reward shaping-based approaches are tightly coupled with their specifically designed individual rewards, limiting their transferability to alternative reward formulations and restricting their practical applicability across diverse multiagent scenarios. This tight coupling further complicates aligning mixed rewards with the team's objective.

The latter category, knowledge transfer-based approaches, focuses on extracting valuable insights from individual policies trained with individual rewards and transferring this knowledge (e.g., policy gradients or policy distributions) to enhance team policy learning. As shown in Fig. 1(b), these methods typically leverage transfer learning techniques [24], [25] to adapt policies trained with individual rewards for fine-tuning target tasks with sparse team rewards. Unfortunately, these approaches still ignore the issue of policy inconsistency between learned policies and optimal team policies during the knowledge transfer process. While multiagent evolutionary reinforcement learning (MERL) [26] and individual reward-assisted team policy learning (IRAT) [23], [27] attempt to address policy inconsistency by sharing replay buffers and enhancing policy similarity between learned policies and optimal team policies, they cannot fully guarantee policy consistency. Here, policy consistency requires that the learned policies achieve the same team performance as optimal team policies. Moreover, a critical oversight in the design of existing approaches is that individual rewards are not universally beneficial for team policy learning; in some cases, they may even directly conflict with team objectives [28]. Consequently, determining the optimal balance between individual and team

reward-driven policy optimization throughout the learning process remains a significant challenge.

Based on the above insights, our research aims to develop an approach that effectively and efficiently leverages individual rewards while addressing three critical objectives: 1) ensuring policy consistency with optimal team policy in terms of gained team rewards; 2) dynamically balancing the tradeoff between team and individual rewards; and 3) maintaining compatibility with diverse individual rewards.

To achieve the above goals, we propose the consistency between learned policy and optimal team policy (CLOT) framework shown in Fig. 1(c). We first cast MARL with individual rewards as a consistency-constrained policy optimization: maximize the mixed return of the learned policy while enforcing that its attained team return matches that of the team-optimal policy. We then solve this problem via Lagrangian dual optimization and prove the equivalence between the formulated problem and its Lagrangian dual problem. Two technical components make the dual solution practical and sample-efficient: 1) a policy approximation scheme that avoids inefficient concurrent sampling from both mixed and team policies; and 2) a Kullback–Leibler (KL) divergence-based objective reformulation that preserves equivalence while simplifying computation. From this formulation, we derive coupled, closed-form policy optimization objectives for the mixed and team policies and an adaptive update of the Lagrange multiplier (λ) that dynamically reweights team versus individual objectives during training.

In summary, CLOT delivers three advantages over existing individual-reward-based MARL methods.

- 1) *Aligning Learned and Optimal Team Policies*: By optimizing under an explicit consistency constraint and its dual, CLOT keeps the learned policy aligned with the team-optimal objective in terms of attained team returns.
- 2) *Dynamic Tradeoff Between Team and Individual Rewards*: An adaptive update of λ reweights team versus individual terms during training, counteracting misalignment when individual rewards hinder progress toward the team-optimal policy.

- 3) *Compatibility With Diverse Individual Rewards*: CLOT treats individual rewards as values entering the optimization, without assuming specific forms or handcrafted shaping, enabling seamless use across heterogeneous MARL tasks.

Finally, we conduct extensive experiments across SMAC, multiagent particle environment (MPE), and Google research football (GRF) benchmarks to systematically evaluate each of the three advantages of our approach compared to state-of-the-art MARL methods. The experimental results validate all three claimed advantages.

II. RELATED WORK

Existing MARL approaches that incorporate individual rewards can be broadly categorized into two main streams based on their utilization of individual rewards: reward shaping-based approaches and knowledge transfer-based approaches.

A. Reward Shaping-Based Approaches

Reward shaping approaches augment team rewards with individual rewards through a mixed reward function, primarily focusing on effective individual reward construction. These approaches can be categorized into three types based on their individual reward formulation strategies.

Expert knowledge-based approaches leverage domain expertise to design individual rewards [9], [10], [11], [12]. For instance, SMAC [29] rewarded enemy elimination and teammate health, while MAPPER [30] decomposed tasks into subtasks with dense rewards. However, acquiring such domain expertise is often impractical or prohibitively expensive in complex real-world environments [31], [32]. **State-space exploration-based approaches** construct individual rewards from observed states. These methods range from basic state-counting [13], [14] and state-prediction [21] to more sophisticated approaches. Recent advances include EDTI/EITI, which focused on influential states [33] and CMBE that addressed diminishing curiosity [34]. LAIES [15] partitioned state spaces to promote external state exploration, while MASER [16] rewarded proximity to target states. Other methods like DIFFER [35] decomposed team experiences into individual components, and population-based approaches [18] guided exploration across different state regions. CMAE [17] and SAME [36] utilized structural knowledge to focus rewards on state subsets. **Action space exploration-based approaches** construct individual rewards to promote diverse actions. CDS [19] maximized mutual information between agent identities and trajectories, while [20] rewarded joint policy diversity between current and previous policies. BDII [37] allocated individual rewards based on the statistical distance between agents' policies, measured by Kullback–Leibler divergence.

While the above studies have shown promising results in addressing reward sparsity through effective individual rewards, the direct introduction of individual rewards can cause the policy learning process to deviate from the team policy's optimization objective. Moreover, most reward shaping-based approaches are tightly coupled with their

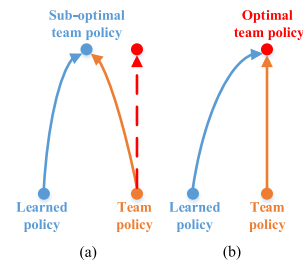


Fig. 2. Comparison of policy convergence targets across different approaches. Blue and orange lines represent learned and team policies, respectively (a) Approaches in [26,27]. (b) Our approach.

elaborately designed individual rewards, limiting their transferability across different individual reward settings.

B. Knowledge Transfer-Based Approaches

Unlike reward-shaping approaches that directly modify reward functions, knowledge transfer approaches focus on leveraging knowledge from individual policies to guide team policy training. These methods commonly employ transferring techniques to incorporate knowledge from individual policies into team policy optimization. For instance, some approaches transferred value functions [24] and action devices [38] from dense individual rewards to facilitate team policy learning [24], [25]. Other methods [39], [40], [41] established multiple critics for different rewards per agent, while updating team policies through value function transfer and decomposition.

In the context of single-agent environments, Chen et al. [28] proposed a two-stage policy optimization method that iteratively updates extrinsic and mixed policies using lower bounds of dual objectives. However, the bias between the real optimization objectives and their approximated lower bounds inevitably affects algorithm performance. Several works have considered policy inconsistency in multiagent settings. Majumdar et al. [26] combined evolutionary algorithms with gradient-based optimization to transfer knowledge from individual policies to team policies through shared replay buffers.

Wang et al. [23] and Li et al. [27] are the most similar approaches to us, like IRAT, which proposed heuristic bidirectional knowledge transmission to enhance policy similarity between individual and team policies. As Fig. 2 shows, IRAT alleviates the policy inconsistency problem but does not guarantee convergence of the learned policy to the optimal team policy. As a result, while blue (learned policy) and orange (team policy) lines gradually approach each other from their starting points and may eventually intersect, the team policy line deviates from its intended direction (red line). Our approach is the first to introduce the consistency policy constraint, enforcing consistency between the mixed policy and the optimal team policy's team rewards. Consequently, the optimization direction of team policy remains unchanged, and the blue and orange lines intersect precisely at the optimal team policy point. Furthermore, [23] and [27] fail to address the dynamic balance between individual and team reward optimization, potentially limiting their effectiveness in scenarios where individual rewards may not consistently benefit team policy learning.

III. PRELIMINARIES

In a cooperative MARL task, the decentralized partially observable Markov decision process (Dec-POMDP) framework defined as $G = \langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}^i\}_{i \in \mathcal{N}}, P, R, \{\mathcal{Z}^i\}_{i \in \mathcal{N}}, \mathcal{O}, \gamma \rangle$ is commonly used to model the problem. Herein, $\mathcal{N} = \{1, 2, \dots, n\}$ denotes the set of agents. $\mathcal{S}$ represents the global state space, with $s \in \mathcal{S}$ denoting the environmental state. $o^i \in \mathcal{Z}^i$ represents the local observation by agent i based on the observation function $\mathcal{O} : \mathcal{S} \times \mathcal{N} \rightarrow \mathcal{Z}^i$, where $\mathcal{Z}^i$ denotes the observation space of agent i . $\mathcal{A}^i$ is the action space of agent i , and $a^i \in \mathcal{A}^i$ denotes the action taken by agent i based on its own policy $\pi_{E+}^i(a^i|\tau^i)$. P denotes the transition probability, specifying the probability of transitioning from state s to state s' under a joint action $\mathbf{a} = (a_1^i, a_2^i, \dots, a_n^i)$. R represents the reward function, with $r^E \in R$ denoting the shared team rewards received by all agents. Given the observation–action history $\tau^i \in \mathcal{T}^i = (\mathcal{Z}^i \times \mathcal{A}^i)$ for agent i , agents learn a collaborative policy $\pi_E = \prod_{i=1}^n \pi_E^i(a^i|\tau^i)$ maximizing the cumulative team reward

$$\max J_E(\pi_E) = \mathbb{E}_{\pi_E} \left[\sum_{t=0}^{\infty} \gamma^t r_t^E \right] \quad (2)$$

where $\gamma \in [0, 1]$ is the discount factor.

Recent studies reveal that the team reward r_t^E is often sparse (i.e., $r_t^E = 0$ for most time steps) in typical multiagent environments, which severely limits the efficiency of cooperative MARL algorithms [3], [4]. To alleviate this sparsity, a common practice is to introduce individual rewards into the Dec-POMDP, so that each agent i receives an additional individual reward r_t^i alongside the shared team reward r_t^E [15], [16]. Under this setting, rather than learning a team policy π_E that maximizes the cumulative team reward, each agent i learns a mixed-reward policy π_{E+}^i that maximizes the mixed reward $\hat{r}_t^i = r_t^i + r_t^E$

$$\max \hat{J}_{E+}^i(\pi_{E+}^i) = \mathbb{E}_{\pi_{E+}^i} \left[\sum_{t=0}^{\infty} \gamma^t \hat{r}_t^i \right]. \quad (3)$$

While introducing individual rewards can significantly improve training efficiency, it can also cause the learned policy to deviate from the team-optimal policy due to the change in the effective objective at convergence [23]. Specifically, under the team-reward-oriented objective, the optimal team policy is $\pi_E^* = \arg \max_{\pi_E} J_E(\pi_E)$ as per (2). However, when the objective shifts to mixed rewards, the resulting optimal mixed policy becomes $\pi_{E+}^* = \prod_{i=1}^n \arg \max_{\pi_{E+}^i} \hat{J}_{E+}^i(\pi_{E+}^i)$ according to (3). In general, π_E^* and π_{E+}^* need not coincide (except under potential-based shaping), leading to a policy inconsistency problem. This motivates our method CLOT, which leverages dense individual rewards for efficient training while explicitly aligning the learned policy with the team-return maximization objective.

IV. METHOD

A. Method Overview

In this section, we propose CLOT, which treats team policy alignment as a hard constraint and solves it via a dual that turns alignment into an adaptive penalty controlled by Lagrangian

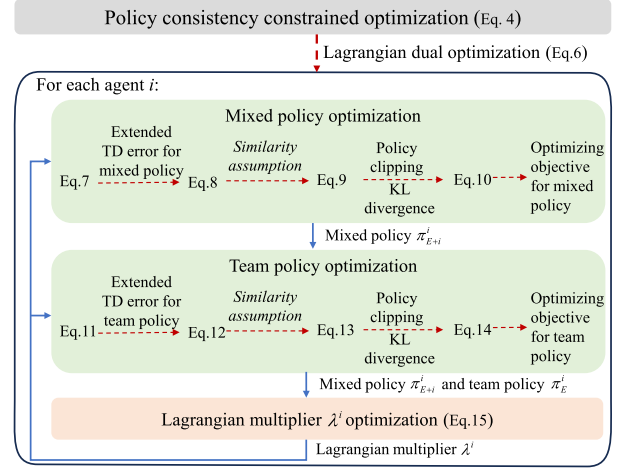


Fig. 3. End-to-end optimization workflow of CLOT. The consistency-constrained objective [see (4)] is dualized [see (6)] with a Lagrange multiplier λ^i . For each agent i , we compute extended TD errors for the mixed and team policies [see (7) and (11)], then form the corresponding objectives for mixed and team policies via the *similarity assumption*, ratio clipping, and a KL regularizer. Finally, λ^i is updated [see (15)]. Solid arrows denote data/optimization flow; dashed arrows denote constraint influences.

multiplier λ . Specifically, as shown in Fig. 3, the workflow of our approach is as follows.

- 1) First, we formulate a consistency-constrained optimization that maximizes mixed returns while aligning team returns with those of the team-optimal policy.
- 2) Second, we apply Lagrangian duality to the formulated problem and prove the equivalence between the primal problem and its Lagrangian dual.
- 3) Third, we derive explicit optimization objectives for the mixed policy π_{E+}^i , the team policy π_E^i , and the Lagrange multiplier λ , further correcting misalignment in individual reward-based MARL.

B. Consistency-Constrained Policy Optimization Model

The research target of this work is to improve the high training efficiency via introducing individual rewards while maintaining the consistency between the learned policy and the optimal team policy. To achieve it, we define the policy optimization problem for each agent i as follows:

$$\begin{aligned} \max_{\pi_{E+}^i} \hat{J}_{E+}^i(\pi_{E+}^i) \\ \text{s.t. } J_E(\pi_{E+}^i) - \max_{\pi_E^i} J_E(\pi_E^i) = 0 \end{aligned} \quad (4)$$

where $J_E(\pi_{E+}^i) = \mathbb{E}_{\pi_{E+}^i} [\sum_{t=0}^{\infty} \gamma^t r_t^E]$ evaluates the cumulative team return gained by mixed policy π_{E+}^i .

In (4), the objective function is to maximize cumulative mixed return $\hat{J}_{E+}^i$, whereas the equality constraint requires the team return of the learned policy to match that of the optimal team policy. Although the cumulative team return depends on the joint policy, during agent i 's optimization, we condition its evaluation on agent i 's local policy and marginalize over teammates' behaviors via a centralized critic within a CTDE paradigm. Thus, other agents' influences are

implicitly captured in the evaluation, while execution uses decentralized actors.

To tackle the intractability of directly solving the policy optimization problem with the unknown term $\max_{\pi_E^i} J_E^i(\pi_E^i)$ in (4), which is transformed into its Lagrangian dual form

$$\min_{\lambda^i} \left[\max_{\pi_{E+i}} \hat{J}_{E+i}(\pi_{E+i}^i) + \lambda^i \left(J_E(\pi_{E+i}^i) - \max_{\pi_E} J_E(\pi_E^i) \right) \right] \quad (5)$$

where λ^i represents the Lagrangian multiplier associated with the consistency constraint for agent i .

To prove the equivalence of (4) and its Lagrangian dual, we make the following assumptions.

Assumption 1: There exists a policy π_{E+i}^i satisfying both $J_E(\pi_{E+i}^i) \geq \max_{\pi_E} J_E(\pi_E^i)$ and $-J_E(\pi_{E+i}^i) \geq -\max_{\pi_E} J_E(\pi_E^i)$, that is, $J_E(\pi_{E+i}^i) = \max_{\pi_E} J_E(\pi_E^i)$.

Assumption 2: Both the mixed reward $\hat{r}_t^i$ and the team reward r_t^E are bounded.

Remark: Assumption 1 ensures feasibility and is reasonable in multiagent RL settings where individual-guided policies can eventually match team-optimal performance [23], [42]. Assumption 2 is standard in multiagent environments (e.g., rewards bounded in $[0, 20]$ in SMAC and $[-1, 1]$ in GRF).

Theorem 1: Under Assumptions 1 and 2, strong duality holds for the optimization problem in (4).

The complete proof is provided in Appendix A1.

C. Dual Formulation and Actionable Objectives

To find the optimal solution of (5), we first rewrite it for each agent i as follows:

$$\min_{\lambda^i} \left[\min_{\pi_E} \max_{\pi_{E+i}} \hat{J}_{E+i}(\pi_{E+i}^i) - \lambda J_E(\pi_E^i) \right] \quad (6)$$

where $\hat{J}_{E+i}(\pi_{E+i}^i) := \hat{J}_{E+i}(\pi_{E+i}^i) + \lambda^i J_E(\pi_{E+i}^i) = E_{\pi_{E+i}^i} \left[\sum_{t=0}^{\infty} \gamma^t [(1 + \lambda^i) r_t^E + r_t^i] \right]$.

From the definition of $\hat{J}_{E+i}(\pi_{E+i}^i)$, the Lagrangian multiplier λ^i serves as a dynamic balancing factor between team and individual rewards in the mixed policy optimization objective for agent i . A larger value of λ^i assigns greater importance to team rewards, while a smaller value emphasizes individual rewards. Next, we will discuss the details of the optimization procedure to optimize the mixed policy, team policy, and λ^i parameter through an iterative optimization method.

Optimizing Mixed Policy: The optimization objective for the mixed policy π_{E+i}^i of agent i in (6) is formulated as $\max_{\pi_{E+i}^i} \hat{J}_{E+i}(\pi_{E+i}^i) - \lambda^i J_E(\pi_E^i)$, which can be further expanded using the performance difference lemma [28], [43] (detailed expansion is provided in Appendix A2)

$$- \sum_{\tau^i \in T^i} d_{\rho_0}^{\pi_{E+i}^i, \gamma}(\tau^i) \sum_{a \in A} \pi_E^i(a_t^i | \tau_t^i) U_{E+i}(\tau_t^i, a_t^i) \quad (7)$$

where $V_{E+i}^{\pi_{E+i}^i, \lambda^i} = \mathbb{E}_{\pi_{E+i}^i} \left[\sum_{t=0}^{\infty} \gamma^t [r_t^i + (1 + \lambda) r_t^E] \right]$ represents the value function of the mixed policy and $d_{\rho_0}^{\pi_{E+i}^i, \gamma}(\tau^i) = \sum_{t=0}^{\infty} \gamma^t P(\tau_t^i = \tau^i | \rho_0, \pi_E^i)$ denotes the discounted state visitation distribution. U_{E+i} as an extended temporal difference (TD) error for evaluating the mixed policy under policy consistency

constraint is defined as

$$U_{E+i} := \lambda r_t^E - V_{E+i}^{\pi_{E+i}^i, \lambda^i}(\tau_t^i) + \gamma V_{E+i}^{\pi_{E+i}^i, \lambda^i}(\tau_{t+1}^i). \quad (8)$$

A significant challenge in directly optimizing policies according to (7) arises from data inefficiency, as it is impractical to simultaneously sample from both π_{E+i}^i and π_E^i . To address this limitation, we introduce *similarity assumption* [28], [43], [44], [45], which approximates the team policy using trajectories from the mixed policy when these two policies remain sufficiently close in their behavioral space. Thus, the optimization objective is simplified as

$$- \sum_{\tau^i \in T^i} d_{\rho_0}^{\pi_{E+i}^i, \gamma}(\tau^i) \sum_{a \in A} \pi_E^i(a_t^i | \tau_t^i) \frac{\pi_{E+i}^i(a_t^i | \tau_t^i)}{\pi_E^i(a_t^i | \tau_t^i)} U_{E+i}(\tau_t^i, a_t^i). \quad (9)$$

To further enhance computational efficiency, we leverage the clipping policy ratio technique and KL divergence between mixed and team policies to simplify the computation on the distribution of policy trajectories in (9). Note that the KL divergence between the mixed and team policies is not added as a heuristic term like MAPPO [44] since the training objective is an algebraically equivalent reformulation via an exact add-and-subtract KL construction. The resulting optimization objective for the mixed policy is expressed as (with comprehensive derivation details provided in Appendix A2)

$$- \mathbb{E}_{\pi_E^i} \left[\left\{ \text{clip} \left(\frac{\pi_{E+i}^i(a_t^i | \tau_t^i)}{\pi_E^i(a_t^i | \tau_t^i)}, 1 - \epsilon, 1 + \epsilon \right) U_{E+i}, \frac{\pi_{E+i}^i(a_t^i | \tau_t^i)}{\pi_E^i(a_t^i | \tau_t^i)} U_{E+i} \right\} - D_{KL}(\pi_{E+i}^i \| \pi_E^i) \right]. \quad (10)$$

Optimizing team policy: The optimization objective for the team policy π_E^i in (6) can be elegantly reformulated as

$$\max_{\pi_E^i} \lambda^i J_E(\pi_E^i) - \hat{J}_{E+i}(\pi_{E+i}^i). \quad (11)$$

Following an approach analogous to the mixed policy optimization, we systematically transform this objective through a series of mathematical derivations (comprehensive details are provided in Appendix A3):

$$\lambda^i J_E(\pi_E^i) - \hat{J}_{E+i}(\pi_{E+i}^i) := - \mathbb{E}_{\pi_{E+i}^i} \left[\sum_{t=0}^{\infty} \gamma^t U_E(\tau_t^i, a_t^i) \right] \quad (12)$$

where $U_E(\tau_t^i, a_t^i) = (1 + \lambda^i) r_t^E + r_t^i - \lambda^i V_{E+i}^{\pi_{E+i}^i}(\tau_t^i) + \sum_{t=0}^{\infty} \gamma^{t+1} \lambda^i V_{E+i}^{\pi_{E+i}^i}(\tau_{t+1}^i)$ represents the extended TD error for team policy evaluation. Based on the *similarity assumption*, the optimization objective is simplified by approximating the mixed policy with team policy as

$$- \sum_{\tau^i \in T^i} d_{\rho_0}^{\pi_{E+i}^i, \gamma}(\tau^i) \sum_{a \in A} \pi_{E+i}^i(a_t^i | \tau_t^i) \frac{\pi_E^i(a_t^i | \tau_t^i)}{\pi_{E+i}^i(a_t^i | \tau_t^i)} U_E(\tau_t^i, a_t^i). \quad (13)$$

Leveraging the same principled approach of policy ratio clipping and introducing KL divergence employed in the

mixed policy optimization, we derive the final optimization objective for the team policy

$$- \mathbb{E}_{\pi_{E+i}^j} \left[\left| \left\{ \text{clip} \left(\frac{\pi_{E+i}^j(a_t^i | \tau_t^i)}{\pi_{E+i}^j(a_t^i | \tau_t^i)}, 1 - \varepsilon, 1 + \varepsilon \right) U_E, \frac{\pi_{E+i}^j(a_t^i | \tau_t^i)}{\pi_{E+i}^j(a_t^i | \tau_t^i)} U_E \right\} \right| - D_{KL}(\pi_{E+i}^j \| \pi_{E+i}^j) \right]. \quad (14)$$

Optimizing Lagrangian Multiplier: To effectively balance individual and team rewards, the Lagrangian multiplier λ^i must be dynamically adjusted throughout the learning process. We employ a principled gradient descent approach to update λ^i , based on the optimization objective defined in (6). The gradient-based update rule for λ^i is formulated as follows (comprehensive derivation details are provided in Appendix A4):

$$\lambda^i \leftarrow \lambda^i - \alpha \mathbb{E}_{\pi_{E+i}^j} \left[\sum_{t=0}^{\infty} \gamma^t \min \left\{ \frac{\pi_{E+i}^j(a_t^i | \tau_t^i)}{\pi_{E+i}^j(a_t^i | \tau_t^i)} A^{\pi_{E+i}^j}(\tau_t^i, a_t^i), \text{clip} \left(\frac{\pi_{E+i}^j(a_t^i | \tau_t^i)}{\pi_{E+i}^j(a_t^i | \tau_t^i)}, 1 - \varepsilon, 1 + \varepsilon \right) A^{\pi_{E+i}^j}(\tau_t^i, a_t^i) \right\} \right] \quad (15)$$

where α represents the step size controlling the magnitude of updates and $A^{\pi_{E+i}^j}(\tau_t^i, a_t^i)$ denotes the standard TD error for team policy evaluation, defined as

$$A^{\pi_{E+i}^j}(\tau_t^i, a_t^i) = r_t^E + \gamma V_{E+i}^{\pi_{E+i}^j}(\tau_{t+1}^i) - V_{E+i}^{\pi_{E+i}^j}(\tau_t^i). \quad (16)$$

D. Implementation and Complexity Analysis

CLOT is implemented within the CTDE framework, with the MAPPO algorithm as the backbone. Both the mixed policy π_{E+i}^j and the team policy π_E^j are independently parameterized by separate neural networks initialized with identical parameters to satisfy the *similarity assumption* for effective policy approximation. Note that such operations are different from the parameter-sharing strategy since the parameter-sharing strategy typically shares policy and value network parameters across homogeneous agents, while our similarity assumption concerns mixed and team policies of the same agent. During the execution, each agent deploys its mixed policy to interact with the environment, while simultaneously recording essential policy information for both mixed and team policies into separate data buffers, denoted as $\mathcal{D}_m$ and $\mathcal{D}_t$, respectively. The training process follows an iterative optimization strategy. The algorithm first optimizes the mixed policy while maintaining the team policy fixed; subsequently, it updates the team policy based on the collectively optimized mixed policies across all agents. Finally, the Lagrangian multiplier is updated. The complete training procedure of the CLOT algorithm is summarized in Algorithm 1. For comprehensive implementation details, refer to Appendix B.

The computational complexity of CLOT during training is $\mathcal{O}(E \cdot (T_1 + 3n \cdot T_2))$, where E is the number of training episodes; T_1 is the time for agent–environment interaction per episode; T_2 is the time for per-agent policy updates per episode; and n is the number of agents. The factor “3” accounts for

Algorithm 1 Training Procedure of the CLOT Algorithm

- 1: For each agent i , initialize parameters θ_{E+i}^j for mixed policy, θ_E^j for team policy, and Lagrangian multiplier λ^i .
- 2: Initialize the learning rate α .
- 3: Initialize empty data buffer $\mathcal{D}_m$ and $\mathcal{D}_t$ for mixed policy and team policy, respectively.
- 4: **while** *training episode* $\leq$ *episode*_{max} **do**
- 5: **for** $step = 1, 2, \dots, step_{max}$ **do**
- 6: Initialize empty trajectories lists $\mathcal{T}_m$ and $\mathcal{T}_t$ for mixed policy and team policy, respectively.
- 7: Generate mixed action, team action from mixed policy and team policy, respectively.
- 8: Interact with the environment with mixed action.
- 9: Compute the extended TD error U_{E+i} and U_E .
- 10: Store state, mixed action, reward, termination information, and U_{E+i} into $\mathcal{T}_m$.
- 11: Store state, team action, reward, termination information, and U_E into $\mathcal{T}_t$.
- 12: Incorporate $\mathcal{T}_m$ and $\mathcal{T}_t$ into $\mathcal{D}_m$ and $\mathcal{D}_t$, respectively.
- 13: **end for**
- 14: Sample training data from $\mathcal{D}_m$.
- 15: Update the mixed policy according to Eq. 10.
- 16: Sample training data from $\mathcal{D}_t$.
- 17: Update the team policy according to the Eq. 14.
- 18: Update Lagrangian multiplier according to Eq. 15.
- 19: **end while**

updating the mixed policy, the team policy, and the Lagrangian multipliers. For comparison, a standard centralized-training method such as MAPPO maintains a single policy and incurs $\mathcal{O}(E \cdot (T_1 + n \cdot T_2))$ training complexity. Although CLOT has a higher nominal update cost, the practical overhead is typically manageable because T_2 (policy-update time) is significantly smaller than T_1 (interaction time) in on-policy settings. For example, on SMAC-MMM2 with 5M training steps on our hardware, CLOT takes 13 h 52 min versus 10 h 36 min for MAPPO, a 31% increase.

V. EXPERIMENTS

To rigorously evaluate the effectiveness of the proposed CLOT algorithm with respect to its three principal advantages: **policy consistency**, **tradeoff team and individual rewards**, and **compatibility with diverse individual reward settings**, we conducted comprehensive experiments across three widely recognized benchmarks for MARL: SMAC [29], MPE [10], and GRF [9]. The experiments are conducted on a computing platform equipped with an AMD Ryzen 9 7950X CPU and an NVIDIA 4090 GPU with 96GB of memory. Complete experimental hyperparameter settings are provided in Appendix C.

A. Validation on Policy Consistency

Policy consistency represents a critical capability for multi-agent systems operating under mixed motivations, ensuring that learned policies remain aligned with collective objectives. To evaluate this aspect of our approach, we compared the winning rates achieved by CLOT against five baselines across

TABLE I

MEDIAN WINNING RATE (%) AND STANDARD DEVIATION (%) OF EXAMINED MARL ALGORITHMS IN 11 MAPS OF THE SMAC ENVIRONMENT UNDER RULE-BASED REWARD SETTING, USING 5 RANDOM SEEDS AND AT MOST 10M TRAINING STEPS

Map	Difficulty	Ours	IRAT	MAPPO	QMIX	MASER	MAVEN	QPLEX	RQN	VGN
2m_vs_1z	Easy	100.0(0.0)	100.0(0.0)	100.0(0.0)	100.0(0.0)	100.0(0.0)	100.0(0.0)	97.2(2.8)	98.9(1.1)	77.6(18.1)
2s3z	Easy	100.0(0.0)	98.9(1.1)	97.2(2.8)	97.2(2.8)	98.1(1.3)	81.6(14.9)	94.9(2.5)	94.4(0.6)	51.4(9.9)
3m	Easy	100.0(0.0)	100.0(0.0)	97.3(2.3)	92.6(5.1)	98.7(1.1)	100.0(0.0)	93.6(1.1)	98.9(1.0)	63.3(5.0)
1c3s5z	Easy	100.0(0.0)	60.9(13.4)	59.3(13.2)	91.4(1.6)	90.2(1.6)	68.4(24.1)	93.8(0.4)	60.4(0.8)	0.0(0.0)
3s_vs_5z	Hard	96.7(1.3)	93.5(4.9)	90.3(4.5)	92.1(3.5)	91.1(5.1)	80.2(8.5)	74.6(5.5)	12.5(21.6)	30.0(19.4)
8m_vs_9m	Hard	93.7(6.2)	56.2(15.6)	83.2(3.1)	58.3(24.0)	67.8(9.2)	85.8(8.5)	36.7(1.9)	32.8(29.8)	26.6(5.5)
5m_vs_6m	Hard	65(6.7)	52.6(20.0)	57.8(8.7)	62.5(6.3)	61.1(17.6)	61.8(4.3)	35.7(1.4)	11.7(6.1)	14.7(7.2)
3s5z	Hard	89.0(9.1)	24.4(22.6)	64.6(8.8)	81.0(12.2)	65.2(10.5)	0.8(0.7)	92.5(0.4)	43.3(4.9)	7.5(9.7)
MMM2	Super Hard	62.5(22.0)	19.1(15.3)	3.6(2.5)	71.8(9.9)	78.6(19.5)	0.0(0.0)	78.4(6.1)	4.1(5.9)	6.4(9.3)
6h_vs_8z	Super Hard	89.0(2.2)	37.5(35.0)	10.9(6.6)	49.8(36.1)	71.1(18.2)	0.0(0.0)	4.3(1.7)	0.0(0.0)	52.2(5.7)
3s5z_vs_3s6z	Super Hard	57.1(35.9)	10.0(0.8)	34.3(20.6)	42.4(49.1)	52.4(49.1)	0.0(0.0)	33.8(13.1)	0.0(0.0)	28.3(24.5)

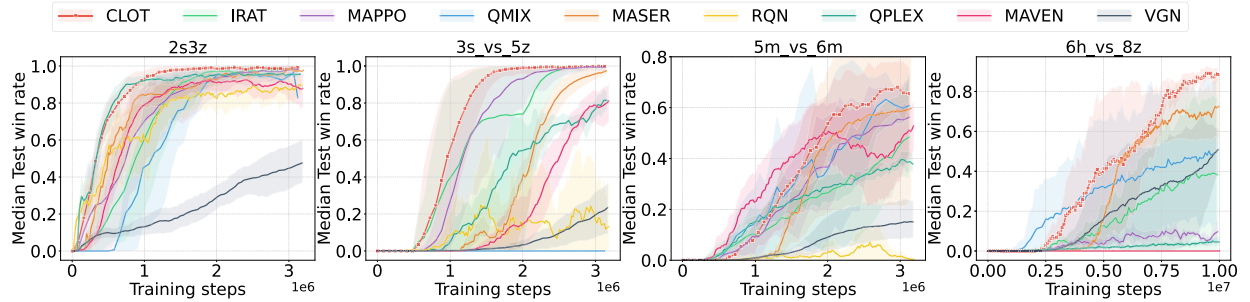


Fig. 4. Training curves of five examined algorithms on selected four maps of SMAC environments. Complete training curves on eleven maps are provided in Appendix C1.

eleven challenging scenarios in the SMAC benchmark, where higher winning rates directly indicate superior team policy performance.

For these experiments, we employed a default reward structure of SMAC (commonly used by most MARL algorithms like MAPPO [8]), consisting of both sparse team rewards and dense individual incentives. Specifically, a sparse team reward of 20 is allocated upon successfully winning a battle, with 0 reward otherwise. Complementing this, dense individual rewards are distributed to each agent based on both team members' health status and enemy eliminations. Agents receive an individual reward of 10 for each defeated enemy unit, along with scaled rewards proportional to their remaining health status throughout the engagement.

We compare CLOT against several representative baselines: IRAT [23], a knowledge transfer-based method that encourages consistency between learned and team policies; MASER [16], a reward shaping method that augments subgoals via individual and global Q -values; MAPPO [8], a widely used clipped-objective MARL algorithm under CTDE; QMIX [46], a value-based method with a mixing network for value factorization; MAVEN [47], a hybrid value and policy-based method that employs a latent variable for temporally extended exploration; QPLEX [48], a state-of-the-art value-based method with a dueling-style factorization of the joint value; RQN [49], a residual Q -learning approach that improves the robustness of factorized value functions; and VGN [50], which leverages graph attention for value decomposition to capture interagent relations. Experiments use SMAC (SC2.4.10).

Experimental results are summarized in Table I. CLOT reaches 100% on all easy maps and attains the highest win

rates on most hard and super hard maps. While IRAT and MASER benefit from utilizing individual rewards and are competitive on several maps, they generally underperform CLOT. These results indicate stronger policy consistency under CLOT, that is, a closer alignment between the learned policy and the optimal team policy. Note that on the Super Hard maps MMM2 and 3s5z_versus_3s6z, the performance variance under our approach is higher at 5M steps, reflecting slower convergence in these coordination-heavy tasks. Extending training to 10M steps reduces the standard deviation from 22.0 to 4.7 (MMM2) and from 35.9 to 18.2 (3s5z_versus_3s6z), suggesting that under-training was the dominant factor.

We further select four representative maps to display the training curves of all examined algorithms in Fig. 4. These visualizations demonstrate that the proposed CLOT algorithm exhibits superior sample efficiency compared to all baseline approaches. CLOT requires significantly fewer training steps to converge, and its converged win rate is either higher than or comparable to the other algorithms across all scenarios. Complete training curves for all 11 maps are provided in Appendix C1.

Additionally, we conducted experiments on two challenging tasks in the GRF environment: 3_versus_1 with the keeper and academy counter-attack (easy). We employ a common setting where the team reward is defined as +1 when the team scores and -1 when the team is scored against. The individual reward is based on the default checkpoint, where an agent receives a reward of 0.1 for possessing the football in a region near the goal. In the GRF environment, we compare the CLOT algorithm with IRAT, MAPPO, and MAPPO (Sparse)

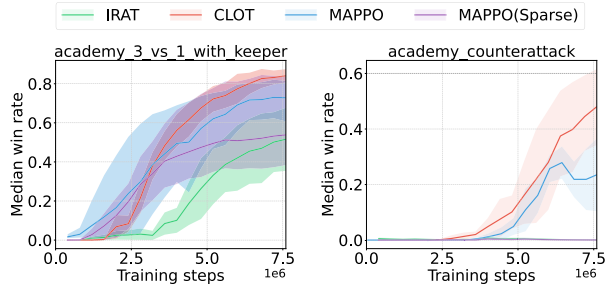


Fig. 5. Experimental results on two maps of the GRF environment.

algorithms. We did not test QMIX or MASER because the GRF environment does not provide global status information.

As shown in Fig. 5, the CLOT algorithm achieves the highest win rate, exceeding 60% on both maps. This result demonstrates that CLOT can deliver superior performance in environments requiring full collaboration among agents, such as passing and off-ball movement. Even in sparse team reward environments, CLOT demonstrates excellent performance in developing effective strategies through the utilization of mixed rewards. These findings further corroborate our earlier observations, demonstrating that CLOT consistently achieves superior team performance compared to all baseline approaches in diverse multiagent environments.

B. Tradeoff Between Team and Individual Rewards

To validate our approach's ability to effectively balance team and individual rewards, we conducted experiments across three MPE environments following the design in [23]. These environments represent distinct scenarios: where individual rewards may: 1) mislead team task completion; 2) conflict with the team task; or 3) facilitate team task completion. Our method's consistent performance across these environments demonstrates its effectiveness in balancing individual and team rewards during policy optimization.

1) *Spread Environment: Misleading Individual Rewards.* In the Spread environment, four agents collaborate to locate two landmarks, receiving a sparse positive reward of 10 when multiple agents simultaneously discover a landmark. Additionally, each agent earns an individual reward based on its minimum distance to undiscovered landmarks. While these individual rewards appear helpful, they may sometimes mislead agents. For instance, when one landmark is discovered, the agent finding it suddenly receives a negative individual reward based on its distance to the remaining undiscovered landmarks. This negative feedback causes agents to question the value of landmark discovery, potentially preventing them from locating the second landmark.

Results in the first column of Fig. 6 show that CLOT achieves the highest team rewards (approximately 10). IRAT and MAPPO follow with rewards around 8, while MAPPO (Sparse) and QMIX perform worst. These findings demonstrate CLOT's ability to identify and overcome suboptimal policy traps caused by misleading individual rewards.

2) *Attack Environment: Conflicting Individual Rewards.* In the Attack environment, three agents collaborate to attack a

single landmark, earning a positive reward of 20 upon success. Each agent incurs penalties based on the distance to the landmark and collisions with other agents (-1 per collision). While individual rewards guide agents toward the landmark, they conflict with team objectives when agents attack the same landmark, as the agents' combined size exceeds the landmark's size. Thus, collision penalties create obstacles to team task completion.

As shown in the second column of Fig. 6, CLOT outperforms all other baselines, achieving team rewards exceeding 20 while mitigating the negative effects of individual rewards. IRAT ranks second with rewards around 7, while MAPPO, QMIX, and MAPPO (Sparse) perform worse with rewards below 5. These results highlight CLOT's superior ability to balance conflicting reward structures.

3) *Predator-Prey Environment: Complementary Individual Rewards.* In the Predator-Prey environment, five predators work together to capture two prey, receiving a sparse positive reward of 20 upon successful capture. Each agent earns an individual reward of 5 when hitting a prey, plus negative rewards based on the distance to the nearest prey. Unlike the previous environments, these individual rewards align with and support team policy learning.

As shown in the third column of Fig. 6, CLOT and IRAT perform similarly, with CLOT achieving slightly higher final team rewards while IRAT performs better during early training. This behavior occurs because, in this environment, individual rewards have a stronger influence on guiding policy updates in alignment with the team rewards.

4) *Large Predator-Prey Environment:* To evaluate CLOT in a larger-scale setting, we conducted experiments on a large predator-prey environment, where both the agent population and the field area are increased by $10\times$ relative to the standard setup, while keeping the reward specification unchanged. The large-scale configuration includes 50 predators and 20 prey.

The experimental results in the fourth column of Fig. 6 show that CLOT achieves the best team performance across five random seeds, demonstrating strong scalability to 50 agents. Meanwhile, the computational overhead remains modest relative to strong baselines. Under our hardware and with 2M training steps in the large predator-prey setting, MAPPO requires 2 days 8 h, IRAT requires 2 days 18 h, and CLOT requires 2 days 21 h. Although CLOT incurs a slightly higher training cost, the overhead is small in practice and is outweighed by the significant performance improvements observed at larger scales.

5) *Adaptive Balancing via Lagrangian Multiplier.* To understand the mechanism behind CLOT's adaptive performance across these diverse environments, we analyze the variation of the Lagrangian multiplier λ during training. The fifth column of Fig. 6 illustrates the temporal evolution of the agent-averaged Lagrangian multiplier λ during training, with results from all three environments of MPE displayed simultaneously for comparison. The results provide insights into how CLOT automatically adjusts the balance between team and individual rewards based on their relationship in each environment.

Notably, λ increases continuously in the Spread and Attack environments, where individual rewards can harm team policy

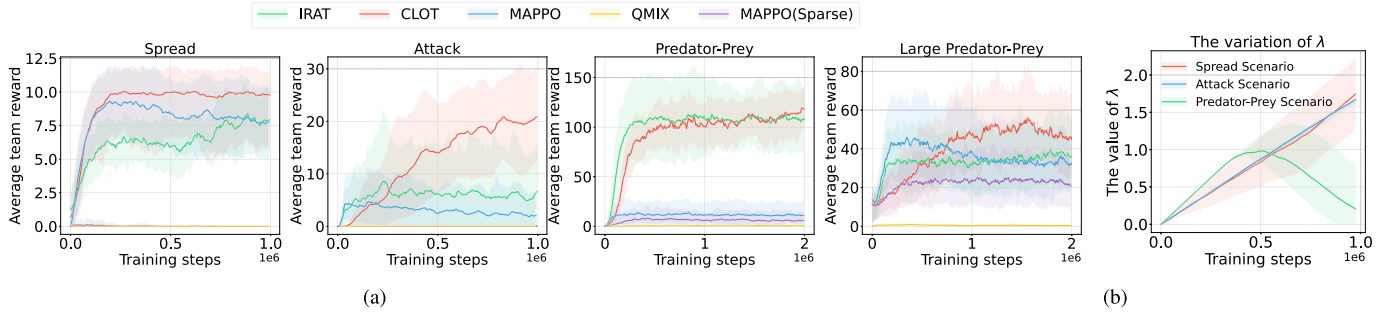


Fig. 6. Experimental results and the variation of Lagrangian multiplier λ on 3 scenarios of MPE across 5 random seeds. (a) Experimental results of team rewards. (b) λ value.

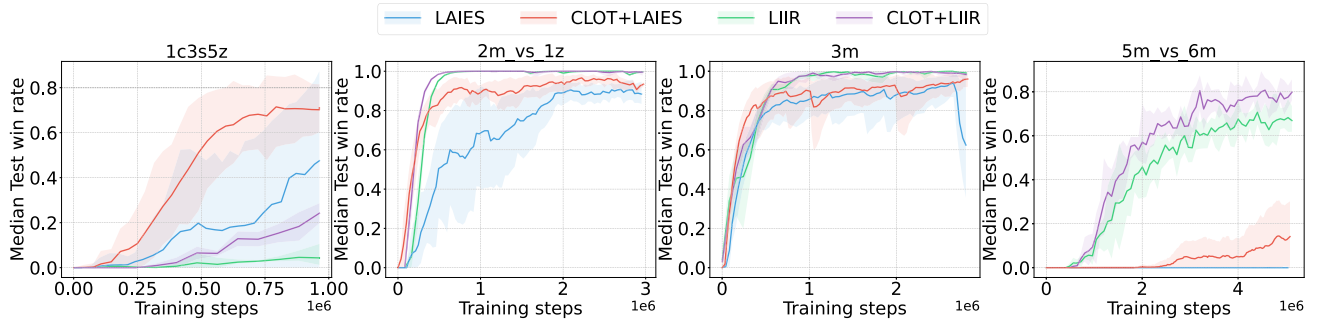


Fig. 7. Training curves of LAIES, LAIES+CLOT, LIIR, and LIIR+CLOT on four SMAC maps using five random seeds.

learning. In contrast, λ first increases, then decreases in the predator-prey environment where individual rewards help team policy learning. As explained in Section IV-C, higher λ values correspond to a greater emphasis on team rewards relative to individual rewards in the optimization objective. The results in the fourth column of Fig. 6 validate our approach’s ability to automatically detect suboptimal traps by increasing the weight of team rewards when necessary (as observed in the Spread and Attack environments), while effectively leveraging individual rewards by reducing the emphasis on team rewards when they align with team objectives (as in the predator-prey environment). Overall, these findings confirm that our method successfully balances team and individual rewards during the learning process, leading to optimal team policies.

C. Validation on Compatibility With Different Individual Rewards Settings

In Section V-A, we validated the effectiveness of CLOT on SMAC using widely adopted handcrafted individual rewards. To further assess CLOT’s compatibility with alternative individual-reward formulations, we integrate CLOT with two representative approaches, LAIES [15] and LIIR [51]. LAIES constructs individual rewards based on the notion of “lazy” agents by estimating the causal effect of actions on external states, whereas LIIR trains a proxy critic to learn an individual reward for each agent by maximizing the agent’s discounted extrinsic return under a mixed-return constraint. We compare each baseline with its CLOT-augmented counterpart, that is, LAIES versus LAIES+CLOT and LIIR versus LIIR+CLOT, to test whether CLOT consistently improves performance across heterogeneous individual-reward designs.

We evaluate on four SMAC maps. For LIIR and LIIR+CLOT, we follow the LIIR setting: a sparse team reward of 20 is given upon a successful win and 0 otherwise; individual rewards are provided by the trained proxy critic with the gradient update in (7) of [51]. For LAIES and LAIES+CLOT, we adopt the configuration like LAIES: the team receives a reward of 1 if all enemies are eliminated, a penalty of -1 if all allies are eliminated, and 0 otherwise. The individual reward of LAIES is computed as

$$r_i^i = \beta_1 r_i^{\text{IDI}} + \beta_2 r_i^{\text{CDI}} \quad (17)$$

where r_i^{IDI} and r_i^{CDI} denote the rewards from individual diligence intrinsic (IDI) motivation and collaborative diligence intrinsic (CDI) motivation for agent i , respectively. The coefficients β_1 and β_2 weight these two components.

Fig. 7 reports the results. Relative to LAIES, LAIES+CLOT yields consistently higher win rates across all four maps. Similarly, relative to LIIR, LIIR+CLOT improves win rate and convergence speed on multiple maps. These findings indicate that CLOT effectively enhances team performance under both heuristic (LAIES) and learned (LIIR) individual-reward designs, supporting its reward-agnostic property.

D. Ablation Studies

It is noteworthy that CLOT incorporates three crucial design elements: the extended TD error, policy approximation based on the *similarity assumption*, and optimization objective reconstruction with KL divergence. To assess the impact of all components on CLOT’s performance, we conduct an ablation study in MPE environments. Specifically, we compare CLOT with its three variants: 1) CLOT-TD(w/o) represents as CLOT

TABLE II

AVERAGE TEAM REWARD OF CLOT AND FIVE VARIANTS, THAT IS, CLOT-TD(w/o), CLOT-SP(w/o), CLOT-KL(w/o), CLOT(3 $\times$) AND CLOT(5 $\times$) ON MPE ENVIRONMENT USING 5 RANDOM SEEDS AND AT MOST 2M TRAINING STEPS

Method	Spread	Attack	Predator-Prey
CLOT	9.8(3.9)	19.7(3.8)	112.5(17.5)
CLOT-TD(w/o)	7.2(2.3)	8.5(4.0)	14.0(6.0)
CLOT-SP(w/o)	9.7(2.5)	13.0(6.0)	106.2(26.5)
CLOT-KL(w/o)	7.5(2.5)	11.0(5.5)	22.5(8.0)
CLOT(3 $\times$)	9.5(2.0)	18.5(6.5)	80.5(22.5)
CLOT(5 $\times$)	12.2(1.7)	17.2(7.5)	107.5(20.0)

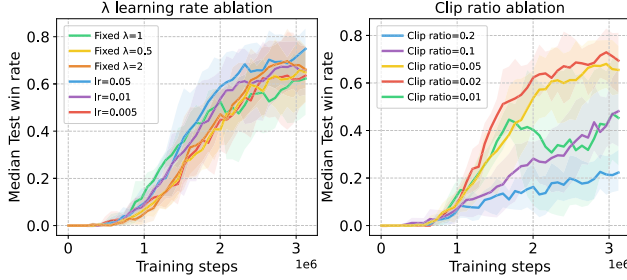


Fig. 8. Ablations on clipping ratios and Lagrangian multiplier learning rates for CLOT on SMAC 5m_versus_6m. Curves report the median test win rate over five random seeds.

with the extended TD error replaced by a standard TD error [defined as (16)]; 2) CLOT-SP(w/o) represents CLOT with the policy approximation removed by initially using distinct parameters between mixed and team policy networks; and 3) CLOT-KL(w/o) represents CLOT without the KL terms-based reconstruction module.

Table II demonstrates that the inclusion of the extended TD error exerts the most significant impact, followed by the KL divergence-based optimization objective reconstruction model, with the policy approximation technique showing the least influence on algorithm performance. The above findings illustrate that the extended TD error, which originates from resolving the policy consistency constraint in the Lagrangian dual problem and the incorporation of mixed rewards, notably contribute to CLOT's enhanced performance. Moreover, the minimal disparity between the outcomes of CLOT-SP(w/o) and CLOT suggests that eliminating the *similarity assumption* does not dramatically affect algorithm performance.

To assess the sensitivity of CLOT to its key hyperparameters, we conduct an ablation on: 1) the convergent clipping ratio for both mixed and team policies and 2) the learning rate of the Lagrangian multiplier. We sweep clipping ratios of 0.20, 0.10, 0.05, 0.02, and 0.01 and Lagrangian learning rates of 0.05, 0.01, and 0.005, and also compare a learned versus a fixed multiplier ($\lambda \in \{0.5, 1, 2\}$). We use the 5m_versus_6m SMAC map with five random seeds and report the median test win rate (Fig. 8).

Across the two panels in Fig. 8, the clipping ratio exerts a stronger influence on performance than the multiplier's learning rate, echoing prior MAPPO findings on the importance of PPO clipping on harder scenarios. Moderate clipping (0.02) achieves the best results and stabilizes training. In contrast, performance is relatively robust to the multiplier's learning rate, with 0.05 performing best on this map. Moreover, the learned multiplier generally matches or outperforms a fixed

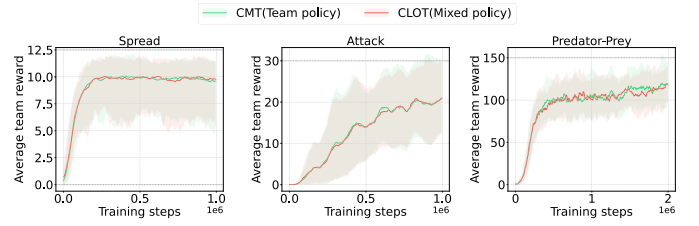


Fig. 9. Comparison between team and mixed policies in terms of gained team rewards.

multiplier. We thus recommend a clipping ratio under 0.02 and a Lagrangian multiplier learning rate under 0.05 for CLOT on SMAC-like tasks.

Moreover, we also investigate the robustness of CLOT with different scaling of individual rewards. In Table II, CLOT(3 $\times$) denotes the CLOT developed with individual rewards amplified three times, while CLOT(5 $\times$) indicates the CLOT developed with individual rewards amplified five times. Comparing CLOT(3 $\times$) and CLOT(5 $\times$) with baseline CLOT across three scenarios, we observe that the performance variation remains within 30%, indicating CLOT's robustness to the scale of individual rewards.

To validate the *similarity assumption* that mixed and team policies remain sufficiently close, we conduct an experiment to compare the team rewards gained by mixed policy and team policies under the three environments of MPE. The results in Fig. 9 show that the mixed and team policies have gained almost the same and equally strong team rewards, validating that the two policies remain sufficiently close.

VI. CONCLUSION AND FUTURE WORK

In this article, we propose CLOT, a novel framework for utilizing individual rewards in MARL environments with sparse team rewards. Unlike existing approaches, CLOT simultaneously achieves three key advantages: 1) ensuring consistency between the learned policy and optimal team policy; 2) effectively balancing individual and team rewards during policy optimization; and 3) maintaining compatibility with diverse individual reward settings. We present a detailed introduction of our approach, which is based on a Lagrangian dual-optimization framework, and validate CLOT's three advantages through extensive experiments across SMAC, MPE, and GRF environments, where it consistently outperforms strong baseline approaches.

Our approach has one main limitation that suggests directions for future work. While we develop policy networks with identical parameters to support the *similarity assumption* during implementation, some divergence between mixed and team policies may still occur during the learning process. Addressing these limitations could strengthen both the theoretical foundations and practical performance of our framework.

APPENDIX

A. Proof and Complete Mathematical Derivation

1) Proof of Strong Duality:

Proof: First of all, we rewrite (4) of the main text in the standard constrained Markov decision process

form

$$\begin{aligned} & \max_{\pi_{E+i}^i} \hat{J}_{E+i}(\pi_{E+i}^i) \\ & \text{s.t. } J_E(\pi_{E+i}^i) \geq \max_{\pi_E^i} J_E(\pi_E^i) \\ & \quad - J_E(\pi_{E+i}^i) \geq -\max_{\pi_E^i} J_E(\pi_E^i). \end{aligned} \quad (18)$$

Although (18) is nonconvex in policy parameters, we establish strong duality by showing that the perturbation function is concave, following the Fenchel-Moreau framework [52] as follows.

Proposition 1 ([52], Corollary 30.2.2): If Assumption 1 holds and the perturbation function of the optimization problem in (18) is concave, then the strong duality holds for (18). Specifically, the perturbation function is defined as follows.

For any $\xi = \{\xi_1, \xi_2\} \in \mathbb{R}^2$, the perturbation function associated with (18) is defined as

$$\begin{aligned} F(\xi) & \triangleq \max_{\pi_{E+i}^i} \hat{J}_{E+i}(\pi_{E+i}^i) \\ & \text{s.t. } J_E(\pi_{E+i}^i) \geq \max_{\pi_E^i} J_E(\pi_E^i) + \xi_1 \\ & \quad - J_E(\pi_{E+i}^i) \geq -\max_{\pi_E^i} J_E(\pi_E^i) + \xi_2. \end{aligned} \quad (19)$$

Proof: Refer to Corollary 30.2.2 of [52]. ■

According to Proposition 1, to prove the strong duality of (18), it suffices that for any $\xi^1 = \{\xi_1^1, \xi_2^1\}, \xi^2 = \{\xi_1^2, \xi_2^2\} \in \mathbb{R}^2$ and $\mu \in (0, 1)$

$$F[\mu\xi^1 + (1-\mu)\xi^2] \geq \mu F(\xi^1) + (1-\mu)F(\xi^2). \quad (20)$$

Suppose that $F(\xi^1)$ and $F(\xi^2)$ are achieved by $\pi_{E+i}^{i,1}$ and $\pi_{E+i}^{i,2}$ (abbreviated as π^1 and π^2), respectively. To establish (20), it needs that for any $\mu \in (0, 1)$, there exists a $\pi_{E+i}^{i,\mu}$ (abbreviated as π^μ), such that $\hat{J}_{E+i}(\pi^\mu) = \mu\hat{J}_{E+i}(\pi^1) + (1-\mu)\hat{J}_{E+i}(\pi^2)$, $J_E(\pi^\mu) \geq \max_{\pi_E^i} J_E(\pi_E^i) + \mu\xi_1^1 + (1-\mu)\xi_1^2$, and $-J_E(\pi^\mu) \geq -\max_{\pi_E^i} J_E(\pi_E^i) + \mu\xi_2^1 + (1-\mu)\xi_2^2$.

To prove the existence of the policy π^μ , we aim to formulate a linear program which is equivalent to Proposition (1) based on a convex and compact occupation measure. Specifically, we first re-write the optimization objective in Proposition (1) as follows:

$$\hat{J}_{E+i}(\pi_{E+i}^i) = \sum_{t=0}^{\infty} \gamma^t \sum_{\tau^i \in T^i} \hat{r}_t^i P(\tau_t = \tau^i | \rho_0, \pi_{E+i}^i) \quad (21)$$

where $P(\tau_t = \tau^i | \rho_0, \pi_{E+i}^i)$ denotes the discounted visitation probability under policy π_{E+i}^i with the initial observation-action distribution ρ_0 .

According to Assumption 2, the mixed reward function is bounded. Therefore, based on the dominated convergence theorem [53], we can exchange the order of two sum in (21) as follows:

$$\hat{J}_{E+i}(\pi_{E+i}^i) = \sum_{\tau^i \in T^i} \hat{r}^i(\tau^i) \sum_{t=0}^{\infty} \gamma^t P(\tau_t = \tau^i | \rho_0, \pi_{E+i}^i). \quad (22)$$

In (22), we attribute a per-visit reward $\hat{r}^i(\tau^i)$ after aggregation over time via the discounted visitation measure.

Similarly, we can reformulate the two constraints in (1) as follows:

$$J_E(\pi_{E+i}^i) = \sum_{\tau^i \in T^i} r^E(\tau^i) \sum_{t=0}^{\infty} \gamma^t P(\tau_t = \tau^i | \rho_0, \pi_{E+i}^i) \quad (23)$$

$$-J_E(\pi_{E+i}^i) = \sum_{\tau^i \in T^i} -r^E(\tau^i) \sum_{t=0}^{\infty} \gamma^t P(\tau_t = \tau^i | \rho_0, \pi_{E+i}^i). \quad (24)$$

By defining the occupation measure $\rho^\pi(\tau^i) = (1-\gamma) \sum_{t=0}^{\infty} \gamma^t P(\tau_t = \tau^i | \rho_0, \pi_{E+i}^i)$, we can define a compact and convex set $\mathcal{R}$ of all occupation measures induced by the policy π_{E+i}^i from Theorem 3.1 in [54] as follows:

$$\mathcal{R} = \left\{ \rho \in \mathcal{M}(T^i) \mid \frac{\rho^\pi(\tau^i)}{1-\gamma} = \sum_{t=0}^{\infty} \gamma^t P(\tau_t = \tau^i | \rho_0, \pi_{E+i}^i) \right\} \quad (25)$$

where $\mathcal{M}(T^i)$ denotes the measures over T^i .

Therefore, we can present a linear program equivalent to (1) on convex set $\mathcal{R}$ as follows:

$$\begin{aligned} F(\xi) & \triangleq \max_{\rho \in \mathcal{R}} \frac{1}{1-\gamma} \sum_{\tau^i \in T^i} \hat{r}^i(\tau^i) \rho(\tau^i) \\ & \text{s.t. } \frac{1}{1-\gamma} \sum_{\tau^i \in T^i} r^E(\tau^i) \rho(\tau^i) \geq \max_{\pi_E^i} J_E(\pi_E^i) + \xi_1 \\ & \quad \frac{1}{1-\gamma} \sum_{\tau^i \in T^i} -r^E(\tau^i) \rho(\tau^i) \geq -\max_{\pi_E^i} J_E(\pi_E^i) + \xi_2. \end{aligned} \quad (26)$$

Denote $\rho_1 \in \mathcal{R}$ and $\rho_2 \in \mathcal{R}$ are the occupation measures associated with π^1 and π^2 , respectively. Because the set $\mathcal{R}$ is convex, there exists an occupation measure ρ_μ such that $\rho_\mu = \mu\rho_1 + (1-\mu)\rho_2 \in \mathcal{R}$. Since both ρ_1 and ρ_2 satisfy the constraints with slacks $\max_{\pi_E^i} J_E(\pi_E^i) + \xi_1$ and $-\max_{\pi_E^i} J_E(\pi_E^i) + \xi_2$ and the sum is linear, ρ_μ satisfies the constraint with slacks $\max_{\pi_E^i} J_E(\pi_E^i) + \mu\xi_1^1 + (1-\mu)\xi_1^2$ and $-\max_{\pi_E^i} J_E(\pi_E^i) + \mu\xi_2^1 + (1-\mu)\xi_2^2$. Therefore, the following inequality holds by using the linearity of summation:

$$\begin{aligned} F[\mu\xi^1 + (1-\mu)\xi^2] & \geq \frac{1}{1-\gamma} \sum_{\tau^i \in T^i} \hat{r}^i(\tau^i) \rho_\mu(\tau^i) \\ & = \mu\hat{J}_{E+i}(\pi^1) + (1-\mu)\hat{J}_{E+i}(\pi^2) \\ & = \mu F(\xi^1) + (1-\mu)F(\xi^2). \end{aligned} \quad (27)$$

Thus, (20) is proved. We can conclude that the strong duality of (18) holds. ■

2) *Optimization Objective of Mixed Policy:* The original optimization objective for mixed policy defined in (6) of the main text is

$$\max_{\pi_{E+i}^i} \hat{J}_{E+i}^{\lambda^i}(\pi_{E+i}^i) - \lambda^i J_E(\pi_E^i). \quad (28)$$

To address the data inefficiency issue in sampling policies π_{E+i} and π_E simultaneously, we make an approximation of optimizing the objective in the training process of the mixed policy network.

First, we expand the objective function as follows:

$$\begin{aligned} & \hat{J}_{E+i}^{\lambda^i}(\pi_{E+i}^i) - \lambda^i J_E(\pi_E^i) \\ &= \mathbb{E}_{\tau_0} \left[V_{E+i}^{\pi_{E+i}^i, \lambda^i}(\tau_0^i) \right] - \lambda^i \mathbb{E}_{\pi_E^i} \left[\sum_{t=0}^{\infty} \gamma^t r_t^E \right] \\ &= -\mathbb{E}_{\pi_E^i} \left[-V_{E+i}^{\pi_{E+i}^i, \lambda^i}(\tau_0^i) + \sum_{t=0}^{\infty} \lambda^i \gamma^t r_t^E \right]. \end{aligned} \quad (29)$$

Since the policy π_E^i has no influence on $V_{E+i}^{\pi_{E+i}^i, \lambda^i}$, we can merge the two components in the second equation in (29), yielding the last equation in (29).

For simplicity, we let $V_t = V_{E+i}^{\pi_{E+i}^i, \lambda^i}(\tau_t^i)$ and $r_t = \lambda^i r_t^E$. We can expand the expression within the expectation in (28) as follows:

$$\begin{aligned} & -V_0 + \sum_{t=0}^{\infty} \gamma^t r_t \\ &= \gamma^0 (r_0 - V_0 + \gamma V_1) + \gamma (r_1 - V_1 + \gamma V_2) + \dots \\ &= \sum_{t=0}^{\infty} \gamma^t (r_t - V_t + \gamma V_{t+1}) \\ &= \sum_{t=0}^{\infty} \gamma^t (\lambda^i r_t^E - V_{E+i}^{\pi_{E+i}^i, \lambda^i}(\tau_t^i) + \gamma V_{E+i}^{\pi_{E+i}^i, \lambda^i}(\tau_{t+1}^i)) \\ &: = \sum_{t=0}^{\infty} \gamma^t U_{E+i}(\tau_t^i, a_t^i). \end{aligned} \quad (30)$$

Therefore, the optimization objective can be rewritten as

$$\begin{aligned} & \hat{J}_{E+i}^{\lambda^i}(\pi_{E+i}^i) - \lambda^i J_E(\pi_E^i) \\ &= -\mathbb{E}_{\pi_E^i} \left[\sum_{t=0}^{\infty} \gamma^t U_{E+i}(\tau_t^i, a_t^i) \right] \\ &= -\sum_{t=0}^{\infty} \sum_{\tau^i \in T^i} \gamma P(\tau_t^i = \tau^i | \rho_0, \pi_E^i) \sum_{a \in A} \pi_E^i(a_t^i | \tau_t^i) U_{E+i}(\tau_t^i, a_t^i) \\ &= -\sum_{\tau^i \in T^i} d_{\rho_0}^{\pi_E^i, \gamma}(\tau^i) \sum_{a \in A} \pi_E^i(a_t^i | \tau_t^i) U_{E+i}(\tau_t^i, a_t^i) \end{aligned} \quad (31)$$

where $d_{\rho_0}^{\pi_E^i, \gamma}(\tau^i) = \sum_{t=0}^{\infty} \gamma^t P(\tau_t^i = \tau^i | \rho_0, \pi_E^i)$ denotes the discounted observation-action frequency under policy π_E^i with the initial observation-action distribution ρ_0 .

To mitigate the data inefficiency resulting from simultaneously sampling policies π_{E+i} and π_E , as per the *similarity assumption*, we substitute π_E^i with π_{E+i}^i and approximate the optimization objectives as follows:

$$\begin{aligned} & \hat{J}_{E+i}^{\lambda^i}(\pi_{E+i}^i) - \lambda^i J_E(\pi_E^i) \\ &= -\sum_{\tau^i \in T^i} d_{\rho_0}^{\pi_{E+i}^i, \gamma}(\tau^i) \sum_{a \in A} \pi_{E+i}^i(a_t^i | \tau_t^i) U_{E+i}(\tau_t^i, a_t^i). \end{aligned} \quad (32)$$

To further simplify the computation process, we introduce the KL divergence between the mixed policy and the team policy while preserving the equality between the original and converted optimization objectives

$$\begin{aligned} & \hat{J}_{E+i}^{\lambda^i}(\pi_{E+i}^i) - \lambda^i J_E(\pi_E^i) \\ &= -\left[\sum_{\tau^i \in T^i} d_{\rho_0}^{\pi_{E+i}^i, \gamma} \sum_{a \in A} \pi_{E+i}^i(a_t^i | \tau_t^i) U_{E+i}(\tau_t^i, a_t^i) \right] \end{aligned}$$

$$\begin{aligned} & + D_{KL}(\pi_{E+i}^i | | \pi_E^i) - D_{KL}(\pi_{E+i}^i | | \pi_E^i) \\ &= -\left[\sum_{\tau^i \in T^i} d_{\rho_0}^{\pi_{E+i}^i, \gamma} \sum_{a \in A} \pi_E^i(a_t^i | \tau_t^i) \frac{\pi_{E+i}^i(a_t^i | \tau_t^i)}{\pi_E^i(a_t^i | \tau_t^i)} U_{E+i}(\tau_t^i, a_t^i) \right] \\ & + D_{KL}(\pi_{E+i}^i | | \pi_E^i) - D_{KL}(\pi_{E+i}^i | | \pi_E^i) \\ &= -\left[\mathbb{E}_{\pi_E^i} \left[\frac{\pi_{E+i}^i(a_t^i | \tau_t^i)}{\pi_E^i(a_t^i | \tau_t^i)} U_{E+i}(\tau_t^i, a_t^i) \right] - D_{KL}(\pi_{E+i}^i | | \pi_E^i) \right] \\ & - D_{KL}(\pi_{E+i}^i | | \pi_E^i). \end{aligned} \quad (33)$$

Leveraging the clip technique presented in [44], we finally convert the objective to

$$\begin{aligned} & \hat{J}_{E+i}^{\lambda^i}(\pi_{E+i}^i) - \lambda^i J_E(\pi_E^i) \\ &= -\mathbb{E}_{\pi_E^i} \left[\min \left\{ \frac{\pi_{E+i}^i(a_t^i | \tau_t^i)}{\pi_E^i(a_t^i | \tau_t^i)} U_{E+i}(\tau_t^i, a_t^i), \right. \right. \\ & \quad \left. \left. \text{clip} \left(\frac{\pi_{E+i}^i(a_t^i | \tau_t^i)}{\pi_E^i(a_t^i | \tau_t^i)}, 1 - \epsilon, 1 + \epsilon \right) U_{E+i}(\tau_t^i, a_t^i) \right\} \right] \\ & - D_{KL}(\pi_{E+i}^i | | \pi_E^i). \end{aligned} \quad (34)$$

The introduction of the KL-divergence term between the mixed policy and team policy in (34) not only assists in acquiring the entire optimization objective but also helps to make the *similarity assumption* valid.

3) *Optimization Objective of Team Policy*: The original optimization objective for the team policy, as defined in (11) of the main text, is

$$\max_{\pi_E^i} \lambda^i J_E(\pi_E^i) - \hat{J}_{E+i}^{\lambda^i}(\pi_{E+i}^i). \quad (35)$$

Similar to optimizing the mixed policy, we rewrite the optimization objective for the team policy

$$\begin{aligned} & \lambda^i J_E(\pi_E^i) - \hat{J}_{E+i}^{\lambda^i}(\pi_{E+i}^i) \\ &= \mathbb{E}_{\tau_0^i} \left[V_E^{\pi_E^i}(\tau_0^i) \right] - \lambda^i \mathbb{E}_{\pi_{E+i}^i} \left[\sum_{t=0}^{\infty} \gamma^t r_t^{E+i} \right] \\ &= -\left[\mathbb{E}_{\tau_0^i} \left[-V_E^{\pi_E^i}(\tau_0^i) \right] + \lambda^i \mathbb{E}_{\pi_{E+i}^i} \left[\sum_{t=0}^{\infty} \gamma^t r_t^{E+i} \right] \right] \\ &= -\mathbb{E}_{\pi_{E+i}^i} \left[-V_E^{\pi_E^i}(\tau_0^i) + \sum_{t=0}^{\infty} \lambda^i \gamma^t r_t^{E+i} \right] \\ &= -\mathbb{E}_{\pi_{E+i}^i} \left[\sum_{t=0}^{\infty} \gamma^t \right. \\ & \quad \left. \left((1 + \lambda^i) r_t^E + r_t^i - \lambda^i V_E^{\pi_E^i}(\tau_t^i) + \gamma \lambda^i V_E^{\pi_E^i}(\tau_{t+1}^i) \right) \right] \\ &: = -\mathbb{E}_{\pi_{E+i}^i} \left[\sum_{t=0}^{\infty} \gamma^t U_E(\tau_t^i, a_t^i) \right]. \end{aligned}$$

By applying the same technique used in optimizing the mixed policy, we obtain the optimization objective for the team policy

$$\begin{aligned} & \lambda^i J_E(\pi_E^i) - \hat{J}_{E+i}^{\lambda^i}(\pi_{E+i}^i) \\ &= -\mathbb{E}_{\pi_{E+i}^i} \left[\min \left\{ \frac{\pi_E^i(a_t^i | \tau_t^i)}{\pi_{E+i}^i(a_t^i | \tau_t^i)} U_E(\tau_t^i, a_t^i), \right. \right. \\ & \quad \left. \left. \text{clip} \left(\frac{\pi_E^i(a_t^i | \tau_t^i)}{\pi_{E+i}^i(a_t^i | \tau_t^i)}, 1 - \epsilon, 1 + \epsilon \right) U_E(\tau_t^i, a_t^i) \right\} \right] \end{aligned}$$

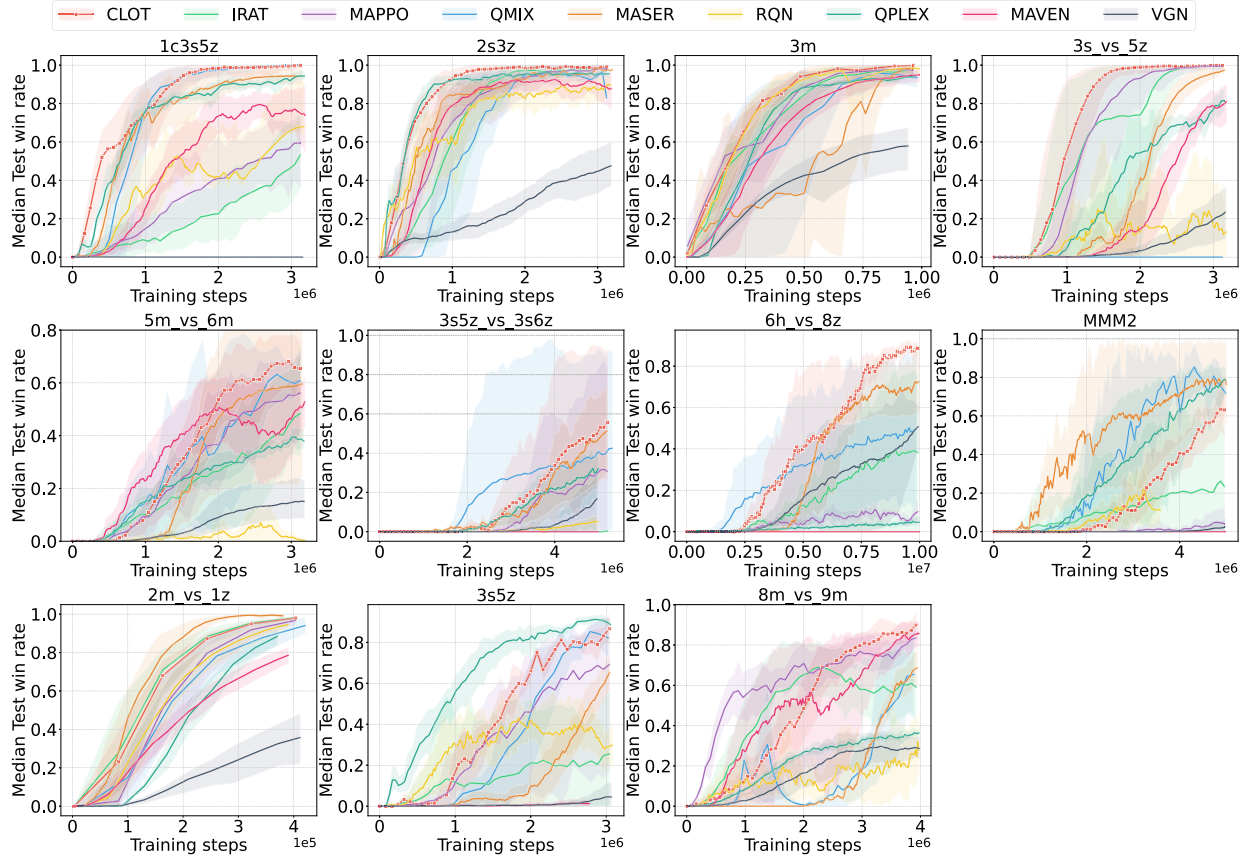


Fig. 10. Training curves of five algorithms on rule-based individual reward setting evaluated on 11 maps of SMAC.

$$-D_{KL}(\pi_E^i || \pi_{E+i}^i).$$

4) *Optimizing the Lagrangian Multiplier λ* : To update Lagrangian multiplier λ^i , we start by deriving the objective with respect to λ^i based on the optimization objective defined in (6) of the main text

$$\nabla_{\lambda^i} (\hat{J}_{E+i}^i(\pi_{E+i}^i) - \lambda^i J_E(\pi_E^i)) = J_E(\pi_{E+i}^i) - J_E(\pi_E^i). \quad (36)$$

To approximate the gradient, we employ a technique presented in the PPO algorithm [44]. The gradient can be lower bounded as follows:

$$\begin{aligned} & J_E(\pi_{E+i}^i) - J_E(\pi_E^i) \\ & \geq \mathbb{E}_{\pi_E^i} \left[\sum_{t=0}^{\infty} \gamma^t \min \left\{ \frac{\pi_{E+i}^i(a_t^i | \tau_t^i)}{\pi_E^i(a_t^i | \tau_t^i)} A^{\pi_E^i}(\tau_t^i, a_t^i), \right. \right. \\ & \quad \left. \left. \text{clip} \left(\frac{\pi_{E+i}^i(a_t^i | \tau_t^i)}{\pi_E^i(a_t^i | \tau_t^i)}, 1 - \epsilon, 1 + \epsilon \right) A^{\pi_E^i}(\tau_t^i, a_t^i) \right\} \right] \end{aligned} \quad (37)$$

where the advantage function $A^{\pi_E^i}(\tau_t^i, a_t^i)$ is defined as $A^{\pi_E^i}(\tau_t^i, a_t^i) = r_t^E + \gamma V_E^{\pi_E^i}(\tau_{t+1}^i) - V_E^{\pi_E^i}(\tau_t^i)$.

Finally, the approximate gradient update step for the Lagrangian multiplier is given by

$$\begin{aligned} \lambda^i & \leftarrow \lambda^i - \alpha \mathbb{E}_{\pi_E^i} \left[\sum_{t=0}^{\infty} \gamma^t \min \left\{ \frac{\pi_{E+i}^i(a_t^i | \tau_t^i)}{\pi_E^i(a_t^i | \tau_t^i)} A^{\pi_E^i}(\tau_t^i, a_t^i), \right. \right. \\ & \quad \left. \left. \text{clip} \left(\frac{\pi_{E+i}^i(a_t^i | \tau_t^i)}{\pi_E^i(a_t^i | \tau_t^i)}, 1 - \epsilon, 1 + \epsilon \right) A^{\pi_E^i}(\tau_t^i, a_t^i) \right\} \right] \end{aligned}$$

with α denoting the step size.

B. Implementation Details

1) *Auxiliary Objectives*: With the MAPPO algorithm as the backbone, CLOT adds an auxiliary objective from MAPPO into the entire optimization objectives of mixed policy and policy, respectively.

When updating the mixed policy, an auxiliary objective from MAPPO is added to maximize $\hat{J}_{E+i}(\pi_{E+i}^i)$ for agent i as follows:

$$\begin{aligned} \max \mathbb{E} \left[\min \left\{ \frac{\pi_{E+i}^i(a_\tau^i | o_\tau^i)}{\pi_{E+i}^{i, \text{old}}(a_\tau^i | o_\tau^i)} A_{E+i}^{i, \text{old}}(a_\tau^i | o_\tau^i), \right. \right. \\ \left. \left. \text{clip} \left(\frac{\pi_{E+i}^i(a_\tau^i | o_\tau^i)}{\pi_{E+i}^{i, \text{old}}(a_\tau^i | o_\tau^i)}, 1 - \epsilon, 1 + \epsilon \right) A_{E+i}^{i, \text{old}}(a_\tau^i | o_\tau^i) \right\} \right] \end{aligned} \quad (38)$$

where $\pi_{E+i}^{i, \text{old}}(a_\tau^i | o_\tau^i)$ denotes the mixed policy of agent i for generating sampling trajectories and $A_{E+i}^{i, \text{old}}(a_\tau^i | o_\tau^i)$ represents the advantage estimate computed under this policy. These terms are critical in the MAPPO framework as they allow the policy update to be stabilized through the clipping mechanism, ensuring controlled and stable learning. Besides, the advantage function for mixed policy $A_{E+i}^{i, \text{old}}(a_\tau^i | o_\tau^i)$ is defined as $A_{E+i}^{i, \text{old}}(a_\tau^i | o_\tau^i) = \hat{r}_t^i + \gamma V_{E+i}^{\pi_{E+i}^{i, \text{old}}}(\tau_{t+1}^i) - V_{E+i}^{\pi_{E+i}^{i, \text{old}}}(\tau_t^i)$.

TABLE III
HYPERPARAMETERS OF CLOT ON SMAC ENVIRONMENTS

Hyper-parameter	Value
Number of fully-connected layers	2
Dim of fully-connected layer	2
Number of GRU layers	1
Dim of RNN hidden layer	64
Optimizer	Adam
Value loss	huber loss
Huber delta	10
Discount factor γ	0.99
Length of episode	400
Activation	ReLU
Use reward normalization	True
Use feature normalization	True
Use agent ID	False
Learning rate of Actor Network	3e-4
Learning rate of Critic Network	3e-4
Initial Policy Clipping Ratio in mixed policy	1.0
Final Policy Clipping Ratio in mixed policy	0.05
Decaying time range of Policy Clipping Ratio	0.4 million training steps
Policy Clipping Ratio in team policy	0.05
Learning Rate of Lagrangian multiplier	0.01

Similarly, when updating the team policy, an auxiliary objective is added to maximize $\hat{J}_E(\pi_E^i)$ for agent i as follows:

$$\max \mathbb{E} \left[\min \left\{ \frac{\pi_E^i(a_\tau^i | o_\tau^i)}{\pi_E^{i,old}(a_\tau^i | o_\tau^i)} A_E^{i,old}(a_\tau^i | o_\tau^i), \right. \right. \\ \left. \left. \text{clip} \left(\frac{\pi_E^i(a_\tau^i | o_\tau^i)}{\pi_E^{i,old}(a_\tau^i | o_\tau^i)}, 1 - \epsilon, 1 + \epsilon \right) A_E^{i,old}(a_\tau^i | o_\tau^i) \right\} \right] \quad (39)$$

where $A_E^{i,old}(a_\tau^i | o_\tau^i) = r_\tau^E + \gamma V_E^{\pi_E^{i,old}}(\tau_{t+1}^i) - V_E^{\pi_E^{i,old}}(\tau_t^i)$.

2) *Update of the Critic Network*: The critic network of the mixed policy for agent i is updated in the direction of minimizing the loss function

$$L(\phi_{E+i}^i) \\ = \mathbb{E} \left[\max \left\{ \left(V_{\phi_{E+i}^i}^i(\tau_t) - R_t^{E+i} \right)^2, \right. \right. \\ \left. \left. \left(\text{clip} \left(V_{\phi_{E+i}^i}^i(\tau_t), V_{\phi_{E+i}^i}^i(\tau_t) - \epsilon, V_{\phi_{E+i}^i}^i(\tau_t) + \epsilon \right) - R_t^{E+i} \right)^2 \right\} \right] \quad (40)$$

where R_t^{E+i} denotes the discounted reward-to-go of agent i 's mixed reward, which is the cumulative reward obtained by agent i in the mixed policy.

The critic network of team policy for agent i is updated in the direction of minimizing the loss function

$$L(\phi_E^i) \\ = \mathbb{E} \left[\max \left\{ \left(V_{\phi_E^i}^i(\tau_t) - R_t^E \right)^2, \right. \right. \\ \left. \left. \left(\text{clip} \left(V_{\phi_E^i}^i(\tau_t), V_{\phi_E^i}^i(\tau_t) - \epsilon, V_{\phi_E^i}^i(\tau_t) + \epsilon \right) - R_t^E \right)^2 \right\} \right] \quad (41)$$

where R_t^E denotes the discounted reward-to-go of agent i 's team reward.

TABLE IV
HYPERPARAMETERS OF CLOT ON THE MPE ENVIRONMENT

Hyper-parameter	Value
Length of episode	25
Activation	ReLU
Use reward normalization	True
Use feature normalization	True
Learning rate of Actor Network	7e-4
Learning rate of Critic Network	7e-4
Initial Policy Clipping Ratio in mixed policy	3.0
Final Policy Clipping Ratio in mixed policy	0.5
Decaying time range of Policy Clipping Ratio	2.0 million training steps
Policy Clipping Ratio in team policy	0.2
Learning Rate of Lagrangian multiplier	0.01

TABLE V
HYPERPARAMETERS OF CLOT ON THE GRF ENVIRONMENT

Hyper-parameter	Value
Initial Policy Clipping Ratio in mixed policy	0.5
Final Policy Clipping Ratio in mixed policy	0.2
Decaying time range of Policy Clipping Ratio	0.5 million training steps
Policy Clipping Ratio in team policy	0.2
Learning Rate of Lagrangian multiplier	0.01

C. Experiment Details

1) *Test on SMAC Environments*: The parameters for the CLOT algorithm on the SMAC environment are provided in Table III. Full experimental training curves of six algorithms, including CLOT, IRAT, MAPPO, QMIX, MASER, and MAPPO(sparse), on 11 maps are provided in Fig. 10.

2) *Test on MPE Environments*: The specific parameters for the CLOT algorithm in the MPE environment are summarized in Table IV.

3) *Test on the GRF Environment*: The parameters for the CLOT algorithm in the GRF environment are summarized in Table V. Some parameter settings have been omitted because they are identical to those used in the MPE environments.

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